Visualising Multi variable functions

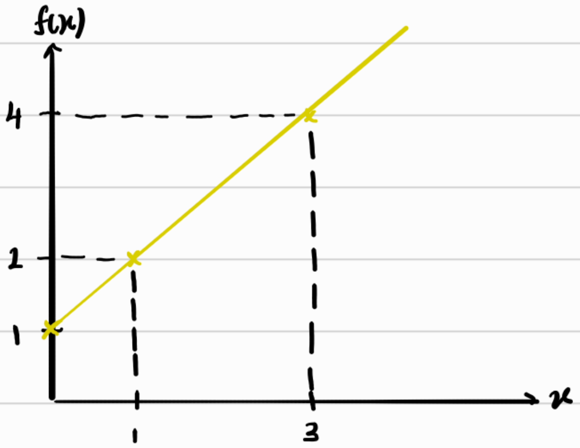
We can create a plot by repeating these 2 steps

1) simply setting the output of the func.
2) finding the inputs

simply set a value

If we want to plot $f(x) = x + 1$, $x = f(x) - 1$

Output	$f(x)$	1	2	4
Input	x	0	1	3



simply set value for this

Similarly we can plot $f(x,y) = 1 + x^2 + y^2$, $x^2 + y^2 = f(x,y) - 1$

$$\underbrace{(x-0)^2}_{a^2} + \underbrace{(y-0)^2}_{b^2} = \underbrace{[f(x,y) - 1]^2}_{c^2}$$

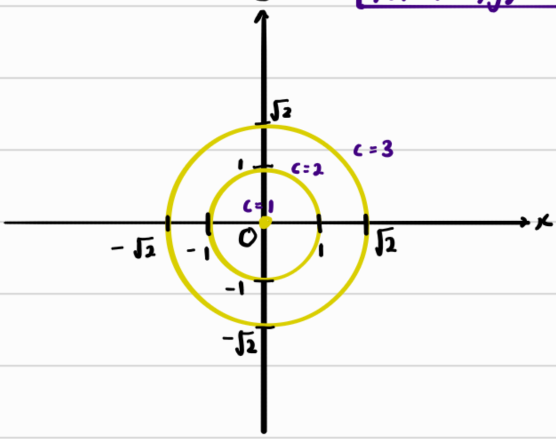
Circle w centre $(0,0)$

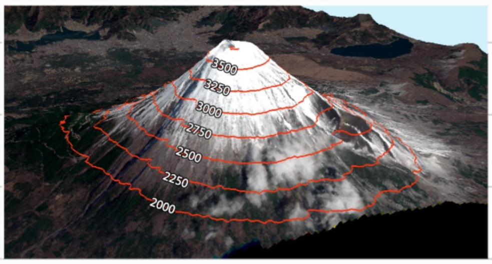
w radius $\sqrt{f(x,y) - 1}$

$f(x,y)$	1	2	3
Radius	0	1	$\sqrt{2}$

This is called an contour

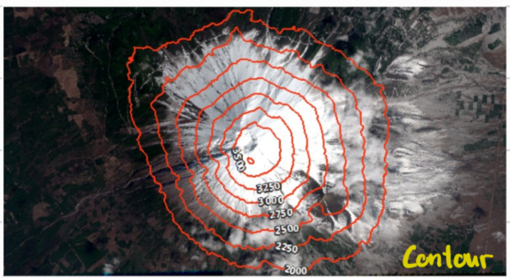
Let $f(x,y) = C$





$f(x,y)$ represents the height of the mountain [eg: 2000, 2250...]

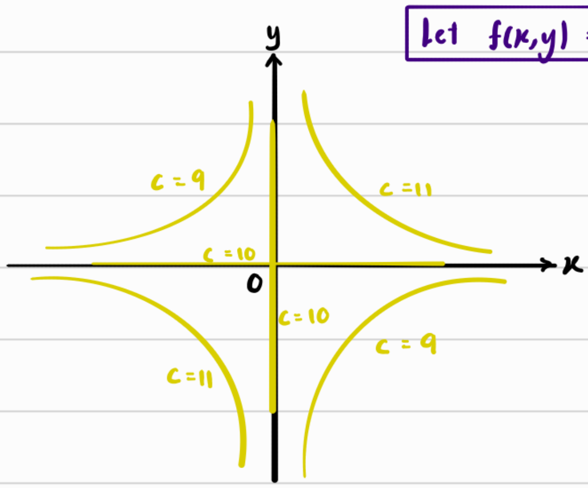
Everytime we set $f(x,y)$ to be a number, we are drawing the contour at that height



Similarly we can plot $f(x,y) = xy + 10$, $y = \frac{f(x,y) - 10}{x}$ Hyperbola , assuming $x \neq 0$

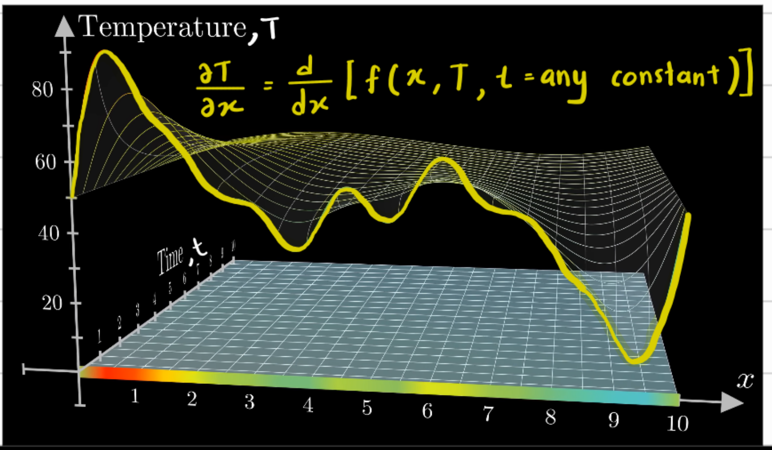
$f(x,y)$	9	10	11
y	$-\frac{1}{x}$	0	$\frac{1}{x}$

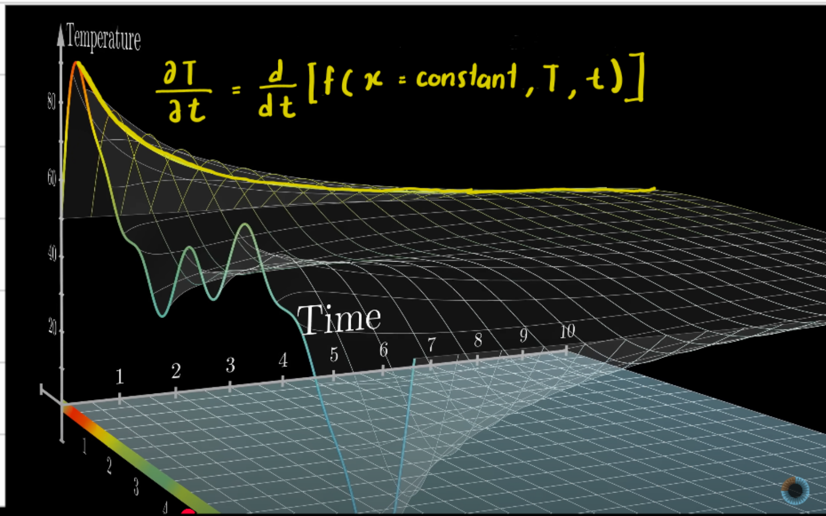
$\boxed{\text{Let } f(x,y) = c}$

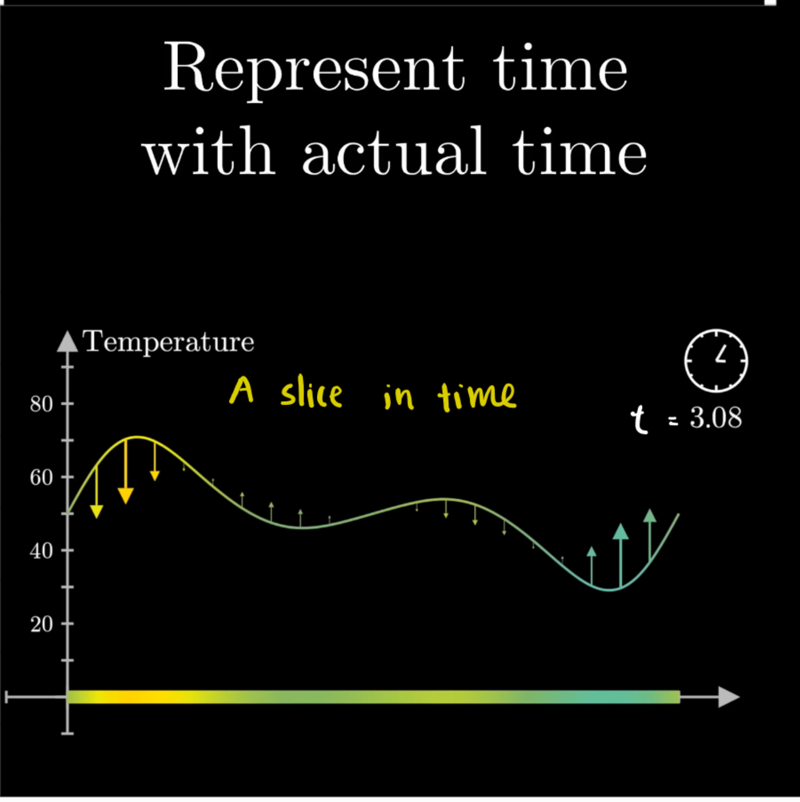


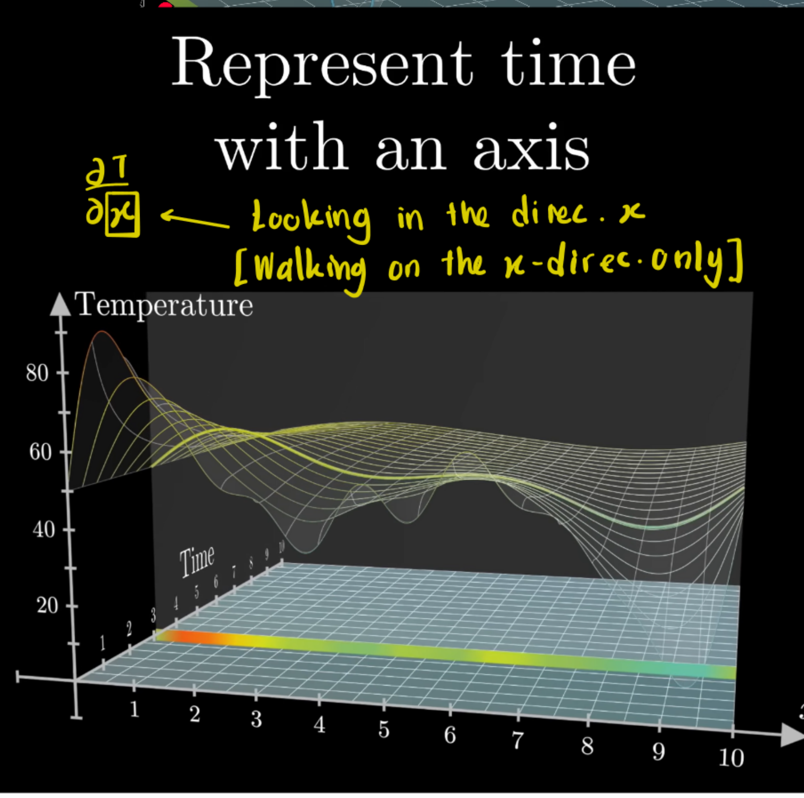
This is called an contour

Partial derivatives









$\frac{\partial f}{\partial \square}$ just means how fast f changes when only $\square$ changes

Partial in terms of x

Example 1 :

$$f(x, y) = 1 + x^2 + y^2$$

Arbitary constant

$$\frac{\partial f}{\partial x} = \frac{d}{dx}[f(x, y = y_0)]$$

$$= \frac{d}{dx}[1 + x^2 + y_0]$$

$$\frac{\partial f}{\partial x} = 2x$$ #

* This means that for any values of y, the slice of that y will have a shape that causes $\frac{df}{dx} = 2x$

Example 2 :

$$f(x,y) = xy$$

$$\frac{\partial f}{\partial x} = \frac{d}{dx}\left[f(x, y = y_0)\right]$$

arbitary constant

$$\frac{\partial f}{\partial x} = \frac{d}{dx}(xy_0)$$

$$= y_0$$

$$\frac{\partial f}{\partial x} = y \; \#$$

Example 3

$$f(x,y) = e^{x^2y}\cos(x+y)$$

$$\frac{\partial f}{\partial x} = \frac{d}{dx}\left[f(x, y_0)\right]$$

$$= \left[\frac{d}{dx}\left(e^{x^2y_0}\right)\right]\cos(x+y_0) + e^{x^2y_0}\left[\frac{d}{dx}\cos(x+y_0)\right]$$

$$= 2xy_0e^{x^2y_0}\cos(x+y_0) + e^{x^2y_0}\left[-\sin(x+y_0)\right]$$

$$\frac{\partial f}{\partial x} = 2xye^{x^2y}\cos(x+y) - e^{x^2+y}\sin(x+y)$$

Don't have to write y_0. Just write y while aknowledging that y is a arbitary constant

Example 4

$$f(x,y) = \frac{1}{xy^2}$$

$$\frac{\partial f}{\partial x} = \frac{1}{y^2}\left[\frac{d}{dx}\left(\frac{1}{x}\right)\right]$$

$$= \frac{1}{y^2}(-x^{-2})$$

$$= -\frac{1}{x^2y^2} \; \#$$

Partial in terms of y

Example 1 :

$$f(x,y) = 1 + x^2 + y^2$$

$$\frac{\partial f}{\partial y} = \frac{d}{dy}\left[1 + x^2 + y^2\right]$$

$$= 2y$$

Example 2 :

$$f(x,y) = xy$$

$$\frac{\partial f}{\partial y} = x$$

Example 3 :

$$f(x,y) = e^{x^2y}\cos(x+y)$$

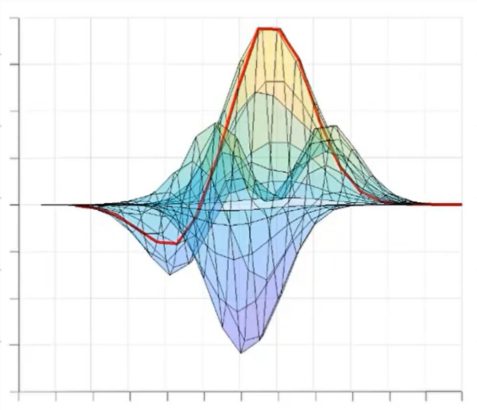
$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(e^{x^2y})\cos(x+y) + e^{x^2y}\frac{\partial}{\partial y}[\cos(x+y)]$$

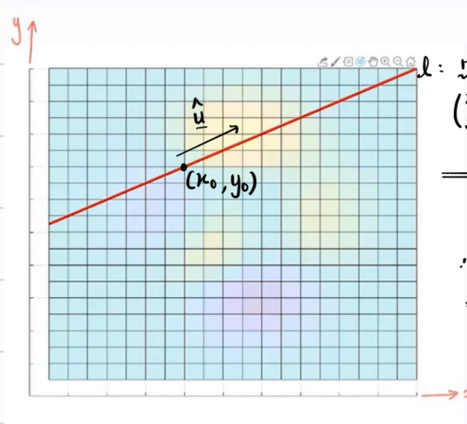
$$= x^2e^{x^2y}\cos(x+y) + e^{x^2y}[-\sin(x+y)]$$

$\boxed{\text{Directional derivative}}$ [Slope at any given line]

What if we want to take a slice in an axis other than x, y, z?

We look at the curve from plan view (top-down view) onto the x-y plane





$l : \underset{\sim}{r} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + u\,\hat{\underline{u}}$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + u\begin{pmatrix} u_x \\ u_y \end{pmatrix}$

$\Longrightarrow$ $\underset{\sim}{x}(u) = x_0 + u\,\hat{u}_x$

$\underset{\sim}{y}(u) = y_0 + u\,\hat{u}_y$

$\therefore f(x,y)$

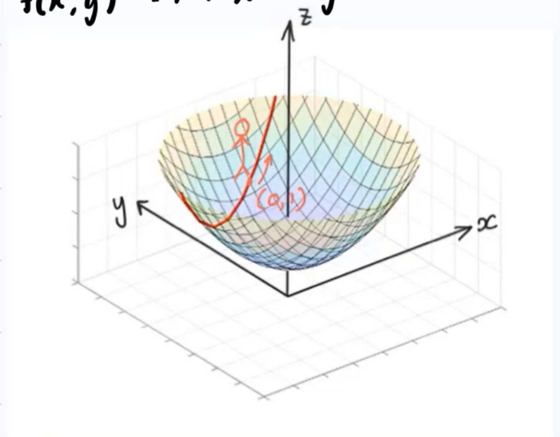
$= f(x_0 + u\hat{u}_x, y_0 + u\hat{u}_y)$

$= F(u)$

Example 1 :

Find the partial derivative of that curve.

$f(x,y) = 1 + x^2 + y^2$



$\therefore f(x,y) = 1 + [x(u)]^2 + [y(u)]^2$

$F(u) = 1 + \left[u\left(\frac{1}{\sqrt5}\right)\right]^2 + \left(1 + u\frac{2}{\sqrt5}\right)^2$

$\frac{\partial F}{\partial u} = 2\left(u\frac{1}{\sqrt5}\right)\frac{1}{\sqrt5} + 2\left(1 + u\frac{2}{\sqrt5}\right)\left(\frac{2}{\sqrt5}\right)$ —— ①

Given $\hat{u} = \frac{1}{\sqrt5}\begin{pmatrix}1\\2\end{pmatrix}$,

$x(u) = x_0 + u\hat{u}_x = 0 + u\left(\frac{1}{\sqrt5}\right)$

$y(u) = y_0 + u\hat{u}_y = 1 + u\left(\frac{2}{\sqrt5}\right)$

Position of stick man

When $u = 0$, $x(u) = x_0 = 0$

$y(u) = y_0 = 1$

$\frac{\partial F}{\partial u} = 0 + 2\left(\frac{2}{\sqrt5}\right)$

$= \frac{4}{\sqrt5}$ #

From previous Example 1's, we have

$\frac{\partial f}{\partial x} = 2x$, $\frac{\partial f}{\partial y} = 2y$

From ①,

$\frac{\partial F}{\partial u} = \left[2\left(u\frac{1}{\sqrt5}\right)\right]\frac{1}{\sqrt5} + \left[2\left(1 + u\frac{2}{\sqrt5}\right)\right]\left(\frac{2}{\sqrt5}\right)$

$= [2x(u)]\,\hat{u}_x + [2y(u)]\,\hat{u}_y$

$$\boxed{\frac{\partial F}{\partial u} = \frac{\partial f}{\partial x}\hat{u}_x + \frac{\partial f}{\partial y}\hat{u}_y}$$

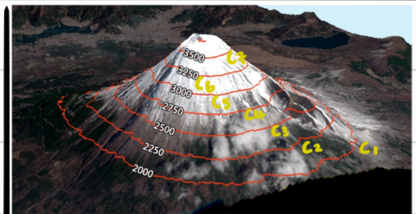
$\Downarrow$

$\frac{\partial F}{\partial u} = \begin{pmatrix}\hat{u}_x\\ \hat{u}_y\end{pmatrix}\cdot\begin{pmatrix}\partial f/\partial x\\ \partial f/\partial y\end{pmatrix}$

$$\boxed{D_u f = \underset{\sim}{\hat{u}} \cdot \nabla f \Big|_{\underset{\sim}{x} = \underset{\sim}{x_0}}}$$

Direc. Derivative

[Rate of change of F in the direc. u]



At a given contour, f(x,y) is constant

$\frac{\partial f}{\partial u} = 0$

$\hat{u} \cdot \nabla f = 0$

$\Longrightarrow \hat{u} \perp \nabla f$

*Note :

1) Direc. Derivative is just a derivative (gradient) at a given direction

2) $\frac{\partial F}{\partial u}$ is max when direc. of $\hat{u}$ is same as ∇f.

Thus, ∇f vector points in the direc. where the function increases the fastest.

3) ∇f is perpendicular to contours

Example 1:

What is the directional derivative of

$$f(x,y) = 1 + x^2 + y^2$$

at $\underline{x} = (0,1)$ in the direction $\frac{1}{\sqrt{5}}\begin{pmatrix}1\\2\end{pmatrix}$

$$\hat{\underline{u}} = \frac{1}{\sqrt{5}}\begin{pmatrix}1\\2\end{pmatrix}$$

$$\underline{\nabla} f = \begin{pmatrix}\frac{\partial f}{\partial x}\\[4pt] \frac{\partial f}{\partial y}\end{pmatrix} = \begin{pmatrix}2x\\2y\end{pmatrix}$$

$$\therefore \ \underline{\nabla} f\Big|_{\underline{x}=(0,1)} = \begin{pmatrix}2\times 0\\2\times 1\end{pmatrix} = \begin{pmatrix}0\\2\end{pmatrix}$$

$$\therefore \ D_u f = \hat{\underline{u}}\cdot\underline{\nabla} f = \frac{1}{\sqrt{5}}\begin{pmatrix}1\\2\end{pmatrix}\cdot\begin{pmatrix}0\\2\end{pmatrix} = \frac{4}{\sqrt{5}} \ \#$$

Example 2:

What is the directional derivative of

$$f(x,y) = e^{x^2}\sin y - y$$

at $\underline{x} = (0, \frac{\pi}{2})$ in the direction $\begin{pmatrix}1\\1\end{pmatrix}$?

$$\hat{\underline{u}} = \frac{1}{\sqrt{1^2+1^2}}\begin{pmatrix}1\\1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$$

$$\underline{\nabla} f = \begin{pmatrix}\frac{\partial f}{\partial x}\\[4pt] \frac{\partial f}{\partial y}\end{pmatrix} = \begin{pmatrix}2x e^{x^2}\sin y\\ e^{x^2}\cos y - 1\end{pmatrix}$$

$$\therefore \ \underline{\nabla} f\Big|_{\underline{x}=(0,\frac{\pi}{2})} = \begin{pmatrix}0\\-1\end{pmatrix}$$

$$\therefore \ D_u f = \hat{\underline{u}}\cdot\underline{\nabla} f = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}\cdot\begin{pmatrix}0\\-1\end{pmatrix} = -\frac{1}{\sqrt{2}}$$

Example 3 :

What is the directional derivative of

$$f(x,y,z) = xyz$$

at $\underline{x} = (1,1,1)$ in the direction $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$?

$$\underline{\hat{u}} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\underline{\nabla} f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix} = \begin{pmatrix} yz \\ xz \\ xy \end{pmatrix}$$

$$\therefore \underline{\nabla} f \Big|_{\underline{x} = (1,1,1)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\therefore D_u f = \underline{\hat{u}} \cdot \underline{\nabla} f = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 1 \quad \#$$

Tangent plane

In single variable calculus, we approx. f(x) using the tangent line eqn

$$f(x) \approx f(x_0) + (x - x_0)[f'(x_0)]$$

Get to the starting position
[Point of tangency]

If we move a little away from x_0, the best guess for f(x) is to go in the direc. of the tangent line

For multivariable calculus, we approx. f(x) using the tangent plane eqn

$$f(x,y) \approx f(x_0, y_0) + [f'_x(x_0, y_0)](x - x_0) + [f'_y(x_0, y_0)](y - y_0)$$

Start position
[Point of tangency]

we move in the direc. of the tangent line of x for a small step $(x - x_0)$

we move in the direc. of the tangent line of y for a small step $(y - y_0)$

f(x,y) is the height of the actual shape

Let z represent the approx. f(x,y).
z represents the height of tangent plane

* $z = f(x,y)$ only at point of tangency.
$z \approx f(x,y)$ near the point of tangency

Rewriting the eqn,

$$f(x,y) \approx f(x_0, y_0) + \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix} \cdot \begin{bmatrix} f'_x \\ f'_y \end{bmatrix}$$

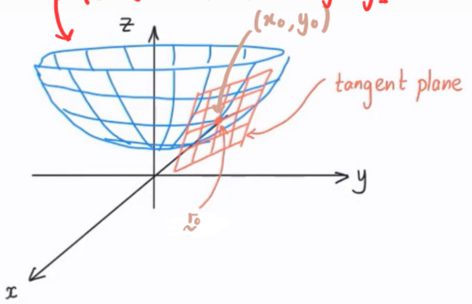
$$\boxed{z = f(x_0, y_0) + [\underset{\sim}{r} - \underset{\sim}{r_0}] \cdot \nabla f|_{r_0 = x_0}}$$

General position vector

Base point.
[Where the tangent plane touches the surface]

Actual shape f(x,y)

[Point of tangency]



* We do not add more terms to the taylor series because we want to find a plane equation. Adding more terms will improve the approx. but cause the plane to curve into the curvature of the 3D image

At (x_0, y_0)

- [Actual Height] Height of shape = [Approx. height] Height of tangent plane

$$f(x,y) = z$$

Near (x_0, y_0)

- [Actual Height] Height of shape ≈ [Approx. height] Height of tangent plane

$$f(x,y) \approx z$$

Summary:

$f(x,y) \rightarrow$ Exact height of actual shape

$z \rightarrow$ Exact height of tangent plane that is a approx. height of actual shape

Example 1 :

Find the tangent plane of $f(x,y) = 1 + x^2 + y^2$ at $\underset{\sim}{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$\nabla f = \begin{pmatrix} 2x \\ 2x \end{pmatrix}$$

$$\nabla f \big|_{\underset{\sim}{r_0} = x_0} = \begin{pmatrix} 2(1) \\ 2(1) \end{pmatrix}$$

$$f(x,y) \approx f(x_0, y_0) + [\underset{\sim}{r} - \underset{\sim}{r_0}] \cdot \nabla f \big|_{\underset{\sim}{r_0} = x_0}$$

$$z = f(x_0, y_0) + [\underset{\sim}{r} - \underset{\sim}{r_0}] \cdot \nabla f \big|_{\underset{\sim}{r_0} = x_0}$$

$$z = f(1,1) + \left[\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right] \cdot \begin{pmatrix} 2(1) \\ 2(1) \end{pmatrix}$$

$$= [1 + 1^2 + 1^2] + \begin{pmatrix} x-1 \\ y-1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$z = -1 + 2x + 2y$$

Output = height = z

Since the approx. was done at $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, we can approx. outputs near $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\Longrightarrow f(1.1, 1) \approx z = -1 + 2(1.1) + 2(1) = 3.2$ [approx. value]

$f(1.1, 1) = 1 + (1.1)^2 + 1^2 = 3.21$ [Actual value]

Check Ans :

When $x = 1$, $y = 1$, $z = f(1,1) = 3$

$\Longrightarrow 3 = -1 + 2(1) + 2(1)$

$3 = 3$ (valid)

Example 2

Find the tangent plane of $f(x,y) = y^2 - xy + 2y + 3x^2$ at $(1, 1, f(1,1))$

$\underline{\nabla} f = \begin{pmatrix} -y + 6x \\ 2y - x + 2 \end{pmatrix}$; $f(1,1) = 1 - 1 + 2 + 3 = 5$

$\underline{\nabla} f \big|_{\underline{r_0} = (1,1,5)} = \begin{pmatrix} -1 + 6 \\ 2 - 1 + 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}$

$f(x,y) \approx f(1,1) + \left[\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right] \cdot \begin{pmatrix} 5 \\ 3 \end{pmatrix}$

$z = 5 + \begin{pmatrix} x-1 \\ y-1 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 3 \end{pmatrix}$

$= 5 + 5x - 5 + 3y - 3$

$z = 5x + 3y - 3$

$5x + 3y - z = 3$ $[ax + by + cz = 0]$

Check ans :

When $x = 1$, $y = 1$, $z = f(1,1) = 5$

$5 = 5(1) + 3(1) - 3$

$5 = 5$ (valid)

Higher order derivatives

For a function $f(x,y)$

① $\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial x}\right]$

② $\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial y}\right]$

③ $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial x}\right]$

④ $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial y}\right]$

} mixed derivative

Example 1:

Find $\frac{\partial^2 f}{\partial y \partial x}$ of $f(x,y) = xy$

$$\frac{\partial f}{\partial x} = y$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial x}\right] = \frac{\partial}{\partial y} y = 1$$

∴ This means as we move in the y-direc., the slope in the x-direc. is increasing at a constant rate of 1.